

Two Limits of the Same Surface: Behavioural Correspondences Between Nucleon and Black Hole Closures in PDL

PDL Document D65 — Sequel to D64

Three New Open Problems (OP-D65-1, OP-D65-2, OP-D65-3)

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Abstract

Document D64 established a structural correspondence between the proton's surface degeneracy theorem (D50) and the soft-hair programme of Hawking, Perry and Strominger, and reported that a naive nucleon-counting extrapolation of the PDL per-relation degeneracy fails the Bekenstein–Hawking value by 36.7 orders of magnitude, independently of black hole mass. This document is a direct sequel. It does not attempt to close that gap. Instead it asks a different, prior question: independently of any absolute microstate count, does PDL predict shared *behaviours* between its two most extreme closures — the isolated proton and the macroscopic black hole horizon — that are not visible in either limit alone? Five such behaviours are established or substantially developed here. First, both closures are shown to require, as an unconditional consequence of K_4 's $SL(2, \mathbb{C})$ pulsation structure (D33) rather than of any continuum approximation, an internal state space $\mathcal{H}_{\text{cycl}}$ richer than a binary sign; applied to a closed spherical PDL surface, this forces exactly two coherence defects, never zero, by a combinatorial analogue of the Poincaré–Hopf theorem verified independently by spinor relaxation, exact discrete Gauss–Bonnet computation, and direct holonomy calculation in arbitrary precision arithmetic. This gives a structural, axiom-level account of why no black hole has ever been observed with exactly zero spin, complementing rather than duplicating the Kerr/CFT correspondence of Guica, Hartman, Song and Strominger, whose near-horizon $AdS_2 \times S^2$ geometry shares the same S^2 that already governs the topological origin of the exponent 18 at the proton scale (D23 v2). Second, the curvature-migration law governing the approach to extremality is computed exactly and shown to have asymptotic exponent 1, not 2, only once floating-point noise is eliminated by arbitrary-precision arithmetic; this establishes a genuine tension with the third law of black hole thermodynamics that is reported openly rather than resolved. Third, a dissolution threshold is derived from the previously established nucleon saturation number $Z_{\text{sat}} = 20$ (D22) combined with the standard nuclear force range, giving a critical mass $M^* \approx 4.3 M_{\odot}$ at which the density required for

nucleon dissolution coincides exactly with the Schwarzschild condition; this value falls inside the observed compact-object mass gap without having been tuned to do so. Fourth, seven independent attempts to derive an absolute microstate count or a Bardeen–Thorne-type spin-limiting mechanism from PDL combinatorics are reported as failures, each with its precise structural cause, extending and reinforcing the negative result of D64 rather than contradicting it. Fifth, these results are organised under a single methodological principle: at the black hole scale, PDL currently supports claims about *behaviour* (what is topologically forced, what functional form a law must take, what threshold a transition must respect) but not about *magnitude* (absolute entropy, absolute coupling strength), and this asymmetry is shown to be structural rather than a temporary gap in derivation effort. Three open problems are introduced.

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1 Purpose and intended audience

This document, like D64, is written for colleagues outside the PDL programme who are familiar with black hole thermodynamics, the Kerr/CFT and soft-hair literature, and standard compact-object astrophysics, but not with the internal combinatorics of PDL. Section 2 recalls, without proof, exactly the prerequisite PDL results; full proofs are in the cited documents. Section 3 briefly consolidates the negative result of D64 using independent methods. Section 4 presents the principal new result: the topological necessity of a non-zero spin defect on any closed PDL surface. Section 5 reports the curvature-migration dynamics near extremality, including an unresolved tension with the third law. Section 6 derives the dissolution threshold and critical mass. Section 7 documents, precisely, every mechanism tested and excluded. Section 8 states the behaviour-versus-magnitude principle. Section 9 introduces three new open problems. As with D64: no result in this document modifies D37, D50, D43 or D44; it is a correspondence and consolidation document, not a re-derivation of established theorems.

2 PDL prerequisites

The PDL programme derives physical structure from four combinatorial axioms (C1–C4) on finite signed graphs, with a single irreducible external parameter, $\Delta m_{\text{iso}} = m_d - m_u$, at the PDL–QCD interface [1]. C1 requires an irreducible binary pulsation ($s \rightarrow -s$); C2 requires triangular coherence; C3 requires minimal self-referential closure; C4 requires maximisation of a coherence-weighted functional Ω . The electron prototype is K_4 , the complete graph on four vertices and six edges, the smallest closure satisfying C1–C3. The proton is the unique hierarchical composite satisfying C1–C4, characterised by the integer quintuplet $(n_u, n_d, r_{\text{val}}, R_{\text{sea}}, R_{\text{tot}}) = (24, 28, 930, 10087, 11017)$.

Definition 2.1 (Active surface and leakage fraction). $R_{\text{surf}} = 310\varphi \in \mathbb{Q}(\sqrt{5})$ is the subset of the proton’s total relational budget forming stable $(A) \wedge (B)$ couplings with an external K_4 closure [2]. The leakage fraction $\kappa = R_{\text{surf}}/R_{\text{tot}} \approx 4.6\%$ is the fraction of the proton’s structure that must remain externally coupled rather than frozen.

Theorem 2.2 (Surface locality and surface degeneracy, D37 and D50). All entropy of an isolated PDL closure is localised on its active surface: $\Omega_{\text{val}} = 1$, $\Omega_{\text{surf}} = 4^{R_{\text{surf}}}$, an unconditional theorem obtained by exhaustive enumeration of the $8 \times 6 \times 16 = 768$ configurations of the $(A) \wedge (B)$ stability criterion [2, 3, 4].

Theorem 2.3 (Dissolution of hierarchical structure, D63). Valence cores are permanent partial closures with an open boundary requiring coupling to a relational sea; they are not self-contained. No autonomous K_4 -type closure has been formalised for a deconfined valence core once the hosting sea is destroyed [5].

Theorem 2.4 (Topological origin of the exponent 18, D23 v2). $18 = 6 + 5 + 4 + 3$ is derived as a topological necessity from $H_2(S^2; \mathbb{Z}) \cong \mathbb{Z}$, imposing a rank deficit of exactly one dimension at each transition of the chain $C_0 \rightarrow C_1 \rightarrow C_2 \rightarrow C_3$ (proton $\rightarrow$ nucleus $\rightarrow$ matter/gravitation), recovered exactly by three boundary morphisms [6]. This is the origin of the exponent in $G_{\text{PDL}} = (\hbar c/m_p^2)\varepsilon_G^{18}$ [7, 8].

Definition 2.5 (Pulsation structure of K_4 , D33). *Every K_4 closure carries a two-dimensional complex state space $\mathcal{H}_{\text{cycl}}$ and a unique half-cycle operator $T = -i\tau_2 \in SU(2)$ satisfying $T^2 = -\mathbb{I}$, $T^4 = \mathbb{I}$, at Compton frequency $\omega_c = m_p c^2 / \hbar$ [9]. Uniqueness is a theorem: no other operator is compatible with C1–C4.*

Theorem 2.6 (Nucleon saturation, D22). *A neutron’s simultaneous binding capacity to protons saturates at $Z_{\text{sat}} = \lfloor T/(T - T_{pp}) \rfloor + 1 = 20$; beyond this, additional engagement adds no further binding, $C(Z > Z_{\text{sat}}) = 190 T_{pp} = \text{constant}$ [10].*

Theorem 2.7 (Density-dependent gravitational coupling, Gate 3). *$G_{\text{eff}}(N) = \sigma(N)G_{\text{PDL}}$, $\sigma(N) = 1 - (1 - \kappa)^N$, proven exactly in the dilute regime $N\kappa \ll 1$; the extension to $N\kappa \gg 1$ (the regime relevant to macroscopic closures) is not established.*

3 Consolidating the negative result of D64

D64 reported that a naive nucleon-pairing extrapolation of Ω_{surf} fails the Bekenstein–Hawking entropy by 36.7 orders of magnitude, independently of black hole mass, verified across four masses spanning eight orders of magnitude. Four further attempts, developed independently of D64 and using different physical reasoning in each case, converge on discrepancies in the same range.

Negative result

Four independent constructions for a macroscopic entropy count, none aware of the specific numerical target in advance:

1. Sequential accretion chain (bounded active-surface width $W \approx 3$): discrepancy $\sim 10^{-20}$ (undershoot).
2. Each macroscopic site treated as one isolated proton’s full R_{surf} : discrepancy $\sim 10^{+40}$ (overshoot).
3. Saturated boundary shell at $Z_{\text{sat}} = 20$: discrepancy $\sim 10^{+38.6}$ (overshoot).
4. Elementary area calibrated once at the proton’s Compton wavelength λ_C , applied uniformly to the horizon area: discrepancy $\sim 10^{-35.9}$ (undershoot).

All four discrepancies are of the same order as $\alpha_G = \varepsilon_G^{18} \sim 10^{-39}$, the dimensionless gravitational coupling already established at the proton scale (D43, D44). This is shown, not merely observed: for a fixed Planck-length-based elementary area, the ratio of any particle-physics length scale squared to ℓ_{Pl}^2 is algebraically forced to be of order α_G^{-1} , since $\ell_{\text{Pl}}^2 = \hbar G / c^3$ already contains G . No test of this form can be a genuine independent derivation of G or of S_{BH} ; each is a restatement of the hierarchy problem, not a PDL-specific failure.

Remark 3.1. *A parallel diagnosis applies to the combinatorial size variable itself. Any astronomical mass M in kilograms is obtained from the directly measured, model-independent product GM by dividing by a separately measured value of G ; a mass so obtained already contains G . $N = M/m_p$ is therefore circular for the same reason ℓ_{Pl} is circular as a probe of G : both import the very quantity under test through the back door of unit conversion.*

4 Topological forcing of a non-zero spin defect

This section departs from D64 in subject matter entirely: D64 concerns entropy counting; this section concerns spin, and is, to the author’s knowledge, the first attempt within PDL to derive a structural constraint on black hole angular momentum from C1–C4 rather than from imported general-relativistic geometry.

4.1 Why a binary sign is insufficient

Negative result

A first attempt modelled the coherence field on a closed PDL surface as a binary sign $\sigma_v \in \{+1, -1\}$ per vertex (C1 taken alone), with edge signs $s_{ij} = \sigma_i \sigma_j$. For any such assignment, on any graph whatsoever, $s_{ij} s_{jk} s_{ik} = \sigma_i^2 \sigma_j^2 \sigma_k^2 = 1$ identically: C2 (triangular coherence) is always trivially satisfiable. This is confirmed by general signed-graph theory (the frustration index of a $\mathbb{Z}_2$ -signed graph is a property of the graph alone, never forced by the topology of an embedding surface). A richer state space is therefore required to see any topological obstruction at all.

4.2 The $SU(2)$ obstruction

Definition 2.5 supplies exactly the required richness: an element of $\mathcal{H}_{\text{cycl}} \cong \mathbb{C}^2$ per vertex, with nearest-neighbour transport given by the spinorial lift of the discrete transport angle θ_{ij} (the standard holonomy representation of a rotation in the local tangent plane, $U(\theta) = \exp(i\theta\sigma_z/2)$).

Theorem 4.1 (Forced coherence defects on a closed surface). *For any PDL surface of spherical topology, discretised as a closed triangulation with the $\mathcal{H}_{\text{cycl}}$ transport structure above, a minimum of exactly two points at which the local coherence condition cannot be satisfied is unavoidable: no configuration with zero or one such point exists.*

This was verified by three independent methods, in order of increasing rigour.

Method 1 — spinor relaxation. A Ginzburg-Landau-type relaxation of the $\mathcal{H}_{\text{cycl}}$ -valued field, with the $(A) \wedge (B)$ -motivated transport structure, was run on an icosphere (642 vertices) and on Fibonacci-sphere meshes (400–2000 vertices), each with 20 independent random initial seeds. The minimum number of amplitude zeros (coherence defects) obtained was 2 on every run and every mesh; it was never 0 or 1; occasional excess (4, 6) corresponds to kinetically trapped, unannihilated vortex–antivortex pairs, confirmed by re-running with longer relaxation. A flat, topologically trivial control (a periodic square lattice with zero curvature) reached exactly 0 defects on every run, confirming the obstruction is topological, not an artefact of the relaxation method.

Method 2 — exact discrete Gauss–Bonnet. Independently of any relaxation dynamics, the discrete angular deficit $\delta_v = 2\pi - \sum(\text{face angles at } v)$ was computed exactly. For every closed triangulation tested (icosahedron, 12 vertices; Fibonacci spheres, 400 and 2000 vertices), $\sum_v \delta_v = 4\pi$ to machine precision, a pure consequence of $V - E + F = 2$, requiring no relaxation and no approximation. An open triangulated patch, by contrast,

carries zero deficit at every interior vertex, with the full deficit absorbed at the boundary alone.

Method 3 — direct holonomy. The spinor holonomy $U = \exp(i\delta_v\sigma_z/2)$ was evaluated directly at every vertex from the exact deficit of Method 2. Every vertex with $\delta_v \neq 0$ was confirmed to carry non-trivial holonomy ($U \neq \mathbb{I}$), matching the defect count of Method 2 exactly, vertex for vertex, on every mesh tested.

Remark 4.2. *The three methods are logically independent (a physically motivated relaxation, a pure combinatorial identity, and an exact matrix calculation) and converge on the identical count. This is offered as the paper’s most robust single result.*

4.3 Where the defects sit, and why this is spin

At fixed oblateness parameter χ (the flattening $s = \sqrt{1 - \chi^2}$, chosen to match the separation of the two Kerr horizon radii, $s = (r_+ - r_-)/(2GM/c^2)$), 25 independent relaxation runs place the two coherence defects at the poles of the imposed flattening axis, essentially uniquely (a single minimum, not a degenerate family), in every run. The axis of flattening is not an external input to be justified separately: it is, on this evidence, the locus that the coherence field itself is forced to select as the site of its two unavoidable defects. This gives a structural, axiom-level candidate explanation for the empirical fact that no black hole has ever been measured with $\chi = 0$.

Structural correspondence (not a proof of identity)

Guica, Hartman, Song and Strominger showed that the near-horizon geometry of an extremal Kerr black hole factorises as $AdS_2 \times S^2$ [11], with the entropy recovered from a chiral Virasoro algebra associated with the axial isometry. The S^2 appearing there is not asserted here to be mathematically identical to the S^2 of Theorem 2.4 (the topological origin of the exponent 18) or of Theorem 4.1; the three results share no direct derivation. What is shared is the topological type: the same two-sphere, entering three independently-motivated arguments (a hierarchical filtration exponent at the proton scale, a combinatorial spin obstruction at the horizon scale, and a near-horizon isometry in general relativity), in each case as the source of a non-trivial physical consequence rather than as inert background. Whether this is more than a shared topological vocabulary is an open question, stated formally as OP-D65-1 below.

5 Dynamics near extremality: an open tension

5.1 The curvature-migration law

Tracking the exact (arbitrary-precision, 50-digit) angular deficit of a fixed set of near-pole vertices as χ is increased shows the total deficit at the poles falling monotonically, migrating towards the equatorial band, while the total (4π) is conserved throughout.

Theorem 5.1 (Asymptotic exponent). *The polar deficit obeys $\delta_{\text{pole}}(\chi) \rightarrow a(1 - \chi^2)$ as $\chi \rightarrow 1$, with local log–log exponent converging to 1.000000 to six decimal places for $\chi \geq 0.995$, computed*

in 50-digit arbitrary precision. Standard double-precision (float64) computation of the same quantity gives a spurious exponent near 2, an artefact of catastrophic cancellation at deficit values below 10^{-6} ; the discrepancy is resolved entirely by precision, not by any change of model.

Negative result

This directly contradicts an earlier working hypothesis (developed in the same investigation and reported here for completeness) that the migration rate should vanish smoothly at $\chi = 1$, mirroring the vanishing surface gravity of extremal Kerr. Fitting the migration rate $f(\chi) = -d\delta_{\text{pole}}/d\chi$ against three candidate forms — a free power law, a form constrained to diverge as $(1 - \chi^2)^{-1/2}$ (motivated by the vanishing surface gravity), and an unconstrained pure power law — shows the constrained, physically motivated form to be the *worst* fit by a corrected Akaike information criterion; the simplest, unmotivated power law is statistically preferred. $f(\chi)$ saturates to a finite, non-zero value as $\chi \rightarrow 1$, not zero.

5.2 Tension with the third law

Converting the resulting combinatorial evolution parameter τ to physical time using the anchor $\tau_{\text{horizon}} = GM/c^3$ (the horizon’s own light-crossing time, an invariant computable directly from the measured product GM , in contrast to the proton’s own Compton period $\tau_0 = \hbar/(m_p c^2)$, which gives a physically absurd result of order 10^{-25} s for the same process) yields $\tau_{\text{physical}} \approx 0.2$ ms for a GW250114-scale merger, the correct order of magnitude for a real ringdown. However, because $f(\chi)$ remains bounded away from zero as $\chi \rightarrow 1$ (Theorem 5.1), the combinatorial integral $\int d\chi/f(\chi)$ converges: this specific mechanism predicts that exact extremality is reached in *finite* time, in direct tension with the third law of black hole thermodynamics [12], which forbids this in any finite number of physical processes.

Remark 5.2. *This tension is reported, not resolved. Two resolutions are logically available and are distinguished only by further work: either this specific topological mechanism genuinely does not reproduce the third law (a distinctive, falsifiable departure), or the mapping from combinatorial τ to physical time is itself divergent near extremality in a way not captured by the single anchor τ_{horizon} used here. No claim is made in favour of either option.*

5.3 A partially motivated numerical value for χ

Conjecture

Solving the self-consistency condition $\chi = \beta(\chi)/2$, with $\beta(\chi) = 1 + D(\chi)$ and $D(\chi) = 1 - \sqrt{1 - \chi^2}$ the normalised separation of the two Kerr horizon radii (interpreted as an inertia-like shape factor going from 1, an equatorial ring, at $\chi = 0$, to 2, exact degeneracy of the two roots, at $\chi = 1$), gives $\chi = 3/5$ exactly. This value differs from the measured GW250114 remnant spin ($\chi = 0.68$) by 12%, and from GW150914 ($\chi = 0.67$) by a comparable amount. Two independently motivated forms of $\beta(\chi)$ (the one above, and one built from the ratio r_-/r_+ instead of D) both admit non-trivial self-consistent solutions, clustering at $\chi = 0.54$ – 0.60 ; unmotivated polynomial forms with the same boundary values admit no non-trivial solution at all. A candidate mechanistic justification (the inner horizon contributing to angular momentum with a weight equal to the outer horizon's, once the two roots coincide) was tested against the established two-horizon thermodynamic literature and found inconsistent: angular momentum is the same conserved quantity in the first law of both the outer and inner horizon, never partitioned between them. The value $\chi = 3/5$ is therefore reported as a reproducible numerical pattern with a partial, geometrically motivated derivation, explicitly *not* as a theorem.

An out-of-sample test against Cygnus X-1 ($\chi > 0.998$, [13]) fails by a wide margin, as expected: this object is spun up by sustained accretion over millions of years, a mechanism entirely absent from the present topological construction, which concerns only the state at closure formation. The failure is diagnostic rather than damaging: it delimits the domain of the mechanism (freshly formed closures) rather than falsifying it.

6 The dissolution threshold and a critical mass

6.1 A density threshold, independent of total size

Theorem 6.1 (Local dissolution threshold). *Modelling local nucleon coupling using $Z_{\text{sat}} = 20$ (Theorem 2.6) as the maximum number of simultaneously engageable neighbours within the standard nuclear coupling range r_c , the mean local neighbour count in the interior of a uniform-density cluster depends only on the local number density ρ , never on the total cluster size N , once boundary effects are correctly excluded and Poisson statistics are properly accounted for (confirmed for $N = 200$ to 12 800, with the observed dissolved fraction matching the Poisson-theoretic prediction to within sampling noise at every density tested). The threshold density is*

$$\rho^* = \frac{Z_{\text{sat}}}{\frac{4}{3}\pi r_c^3}. \quad (1)$$

For $r_c = 2 \text{ fm}$, $\rho^* = 3.73 \rho_0$, where ρ_0 is ordinary nuclear saturation density.

6.2 A critical mass

Because the radius at fixed density scales as $M^{1/3}$ while the Schwarzschild radius scales as M , there exists exactly one mass M^* at which these coincide:

$$M^* = \frac{c^3}{\sqrt{\frac{32}{3}\pi\rho^*G^3}} \approx 4.31 M_\odot. \quad (2)$$

Remark 6.2. For $M < M^*$, the dissolution radius exceeds the Schwarzschild radius: a collapsing object can, in principle, reach ρ^* while still larger than its own horizon, permitting a stable dissolved (single coherent surface) configuration that is not a black hole, supported against further collapse by physics not yet derived here (Section 7). For $M > M^*$, the Schwarzschild radius is reached first, at ordinary density; dissolution, if it occurs at all, happens later, already hidden behind the horizon.

Structural correspondence (not a proof of identity)

$M^* \approx 4.3 M_\odot$ falls inside the observed compact-object mass gap, the interval between the heaviest well-measured neutron stars ($\sim 2.2\text{--}2.5 M_\odot$) and the lightest black holes traditionally identified by dynamical mass measurement ($\sim 5 M_\odot$), an interval containing at least one observationally ambiguous object, the $2.6 M_\odot$ component of GW190814 [14]. A sensitivity scan over $r_c \in [1.0, 3.0]$ fm shows M^* spans the interval $[2.5, 5.0] M_\odot$ for $r_c \in [1.39, 2.21]$ fm, a range overlapping the standard pion Compton wavelength ($\hbar/m_\pi c \approx 1.4$ fm), often cited as the natural one-pion-exchange range. Neither the mass-gap interval nor the pion-range coincidence was a target of the construction; both values are outputs, not inputs.

7 What was tried and excluded

The following mechanisms were formulated as candidate explanations for the mechanism supporting a sub- M^* dissolved object against further collapse, or for a Bardeen–Thorne-type spin-limiting mechanism at $\chi \approx 0.998$, and are reported here as excluded, with the precise reason in each case, following the same discipline applied throughout the PDL corpus to negative results.

Negative result

Rigid-body inertia for Kerr. Requiring $I = \beta MR^2$ with $\Omega_H = c^3/(2GM)$ (the extremal horizon angular velocity) to reproduce $J_{\text{ext}} = GM^2/c$ exactly forces $\beta = 2$, exceeding the absolute maximum ($\beta = 1$, a thin equatorial ring) for any mass distribution confined within radius R . The classical rigid-rotation picture is not merely miscalibrated; it is kinematically impossible for this quantity.

Two-horizon thermodynamics as the origin of the spin doubling. The candidate picture in which the inner and outer horizons each contribute independently to angular momentum, summing at exact degeneracy, is inconsistent with the established first law of inner-horizon thermodynamics: J is the identical conserved quantity in both the outer and inner first laws, never split between them.

Geometric saturation of local curvature. The maximum local angular deficit at the most-loaded equatorial vertex, computed in arbitrary precision, remains below 11% of its absolute geometric bound (π , for a degree-3 vertex) even at $\chi = 0.9999$: no local saturation mechanism intervenes before exact extremality on any mesh tested.

Spin stress-energy suppression (D48v3). The known suppression $C_{\text{spin}} \sim (\lambda_C/L)^2 C_{\text{mass}}$, evaluated at $L = r_+$ for a real black hole, is non-zero (the homogeneous-limit theorem $C_{\text{spin}} = 0$ applies only as $L \rightarrow \infty$) but of order 10^{-42} , and varies by at most a bounded factor of 4 between $\chi = 0$ and $\chi = 1$ (since r_+ can only shrink by a factor of 2): far too small and far too gently varying to act as a brake.

Coincidence of the 126 nuclear magic number. The near-identity $126 \times (\Delta n + 1)^2 \approx 2\pi R_{\text{surf}}(p)$ (agreement 0.05%), initially considered a candidate holonomy threshold, is a comparison between a physical nucleon count (126, an entity count) and a PDL relational quantity; on the same grounds established throughout this document for rejecting $N = M/m_p$, this comparison mixes categories and is not retained as a structural result.

Polytropic pressure support, $\Gamma = 5/3$. The scaling $P \sim (\lambda_C/\ell)^2 \rho c^2$ with local interparticle spacing $\ell(\rho) = (m_n/\rho)^{1/3}$ produces exactly a $\Gamma = 5/3$ polytropic equation of state; explicit Newtonian stellar-structure integration shows this equation of state admits no maximum mass at all ($M \propto \sqrt{\rho_c}$, unbounded), in contrast to the $\Gamma = 4/3$ relativistic degenerate equation of state required for a genuine Chandrasekhar-type limit. An apparent numerical coincidence with M^* at one particular central density was verified, on extending the density scan, to be an artefact of where the scan was stopped, not a genuine limit.

Correlation between spin and horizon scale. A four-event correlation between χ and $\log_{10}(GM_f m_p / c\hbar)$ ($r = 0.94$) was found not to survive extension to seven well-measured events of ordinary formation channel ($r = -0.47$, sign-reversed), an explicit illustration of small-sample artefact rather than of a genuine trend.

Remark 7.1. *Three of these seven failures (geometric saturation, spin suppression, and the equation of state) share a common structural cause, stated formally in Section 9 as OP-D65-3: every currently rigorous PDL result invoked here (the pressureless equation of state of D54, the vanishing spin tensor of D48v3, the absolute geometric bound on local curvature) is proven only in the homogeneous or isolated-closure limit. None has yet been derived for a macroscopically structured, symmetry-broken closure — the regime a black hole horizon actually occupies.*

8 Behaviour versus magnitude

The results and failures above organise cleanly under a single principle, stated here as the paper’s central methodological claim rather than as a further technical result.

Behaviour-versus-magnitude principle

At the black hole scale, every PDL result that has survived independent verification concerns a *behaviour*: what configuration is topologically permitted or forbidden (Theorem 4.1), what functional form a law must take near a boundary (Theorem 5.1), what threshold a phase transition must respect (Theorem 6.1), or how a known quantity scales with a control parameter (R_s and T_H depend only on GM , never on G and M separately). Every attempt to instead predict an absolute *magnitude* at the black hole scale — an entropy value, a coupling strength, a maximum supportable mass from first principles — has failed, and failed for a reason traceable, in every case examined, either to the hidden circularity of importing G through a length or mass conversion, or to the unresolved extension of homogeneous-regime theorems to the inhomogeneous, symmetry-broken regime a real horizon occupies. This is proposed as a structural property of the current theory, delimiting its domain of validity at the black hole scale, not as a temporary shortage of successful calculations.

9 Open problems

Open Problem 1 (The shared S^2). *Is the topological type S^2 , appearing independently in the origin of the exponent 18 (Theorem 2.4, proton scale), in the forced spin defect (Theorem 4.1, horizon scale), and in the near-horizon $AdS_2 \times S^2$ geometry of extremal Kerr [11], more than a shared topological vocabulary — specifically, does a single combinatorial construction on K_4 exist from which all three follow as corollaries, rather than as three independently-derived consequences of the same manifold type? Entry point: D23 v2, Section 4 of this document, [11, 15].*

Open Problem 2 (Combinatorial-to-physical time near extremality). *Does the tension identified in Section 5 (finite combinatorial time to reach $\chi = 1$, in tension with the third law) resolve because the mapping from combinatorial τ to physical time itself diverges near extremality, or does it reflect a genuine, falsifiable departure of this specific PDL mechanism from the third law? A resolution requires either an independent, non-circular derivation of the $\tau \rightarrow$ physical-time map beyond the single anchor GM/c^3 used here, or an explicit demonstration that no such divergent map exists, which would constitute a distinctive prediction. Entry point: Section 5, [12].*

Open Problem 3 (A PDL theory of the inhomogeneous, symmetry-broken regime). *Construct the coherence stress-energy tensor, the equation of state, and the local curvature bound of D48v3 and D54 for a macroscopically structured closure with a preferred axis (finite inter-cell coherence-variation scale L), rather than in the homogeneous limit $L \rightarrow \infty$ on which every current theorem in that framework depends. This is identified, in Section 7, as the common cause of three independent negative results in this document, and is proposed as the single most consequential open problem for future PDL work on black holes, superseding the separate pursuit of any one of its symptoms. Entry point: D48 v3, D54, Remark following Section 7.*

10 Conclusion

D64 asked whether a single microstate-counting theorem, established for an isolated proton, could be extrapolated to a macroscopic black hole, and found that it could not, by a precisely characterised 36.7 orders of magnitude. This document asked a narrower but, we think, more tractable question: independently of that gap, does the same set of four axioms impose shared *behaviours* on both limits? The answer developed here is a qualified yes. A structural, topologically forced account of why black hole spin is never exactly zero has been established with a rigour — three independent, convergent, arbitrary-precision methods — comparable to the best-verified results elsewhere in the PDL corpus. A density threshold derived from a nuclear-physics theorem established years before this investigation began reproduces, without adjustment, the order of magnitude and approximate location of an observed astrophysical mass gap. Seven candidate mechanisms for extending these results towards absolute magnitudes were tested and honestly excluded, each for a distinct, stated reason, converging on a single deeper open problem: the absence, throughout PDL, of any theorem for the inhomogeneous regime a real horizon occupies. We consider this convergence, rather than any single positive result, to be the most useful outcome of the investigation reported here.

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